

APPLICATION LAYER

Introduction

The **Application Layer** is the topmost layer of both the **OSI Model (Layer 7)** and the **TCP/IP Model**.

It provides services directly to end users and application programs.

The application layer acts as an interface between network services and user applications.

Examples:

- Web browsing
- Email
- File transfer
- Remote login
- Online gaming
- Video streaming

Whenever a user accesses a network service, the application layer is involved.

What is an Application?

An application is a software program that performs specific tasks for users.

Examples:

Application	Purpose
Chrome	Web browsing
Firefox	Web browsing
Gmail	Email service
Outlook	Email client
WhatsApp	Messaging
Zoom	Video conferencing
YouTube	Video streaming

The application layer allows these applications to communicate over a network.

Functions of Application Layer

The application layer performs several important functions.

Function	Description
Network Virtual Terminal	Allows remote login
File Transfer	Transfers files between devices
Email Services	Sends and receives emails
Web Services	Provides web access
Directory Services	Provides name and address lookup
Resource Sharing	Shares printers, files, databases
Authentication	Verifies user identity

Services Provided by Application Layer

1. Network Virtual Terminal

Allows a user to access and control a remote computer.

Example:

- SSH
- Telnet

A user sitting in one location can log into another computer through a network.

2. File Transfer Service

Used for transferring files between systems.

Examples:

- FTP
- TFTP
- SFTP

Operations include:

- Uploading files
 - Downloading files
 - Renaming files
 - Deleting files
-

3. Email Service

Allows users to send and receive electronic mail.

Protocols used:

- SMTP
- POP3
- IMAP

Examples:

- Gmail
 - Yahoo Mail
 - Outlook
-

4. Web Service

Provides access to websites and web applications.

Protocols:

- HTTP
- HTTPS

Examples:

- Google
 - Amazon
 - Facebook
-

5. Directory Service

Maps names to addresses.

Example:

Instead of remembering:

142.250.183.78

we use:

www.google.com

This conversion is done by DNS.

Application Layer Protocols

Important application layer protocols include:

Protocol	Full Form	Port
HTTP	HyperText Transfer Protocol	80
HTTPS	HyperText Transfer Protocol Secure	443
FTP	File Transfer Protocol	21
TFTP	Trivial File Transfer Protocol	69
SMTP	Simple Mail Transfer Protocol	25
POP3	Post Office Protocol Version 3	110
IMAP	Internet Message Access Protocol	143
DNS	Domain Name System	53
DHCP	Dynamic Host Configuration Protocol	67/68
Telnet	Telecommunication Network	23
SSH	Secure Shell	22
SNMP	Simple Network Management Protocol	161
NTP	Network Time Protocol	123

HTTP (HyperText Transfer Protocol)

HTTP is the standard protocol used for communication between web browsers and web servers.

Features

- Client-server architecture
- Stateless protocol
- Request-response protocol
- Uses TCP
- Default port: 80

Working of HTTP

Step 1:

Browser sends HTTP request.

Step 2:

Server processes request.

Step 3:

Server sends HTTP response.

Step 4:

Browser displays webpage.

Example

User enters:

www.google.com

Browser sends HTTP request.

Server sends webpage data.

Browser displays webpage.

HTTPS (HyperText Transfer Protocol Secure)

HTTPS is the secure version of HTTP.

It uses:

- SSL (Secure Socket Layer)
- TLS (Transport Layer Security)

Features

- Encryption
- Authentication
- Data Integrity
- Secure communication

Default Port

443

Advantages

- Prevents hacking
- Protects passwords
- Secures online payments
- Ensures privacy

Difference Between HTTP and HTTPS

Feature	HTTP	HTTPS
Security	No encryption	Encrypted
Port	80	443
SSL/TLS	Not used	Used
Data Protection	Low	High
Browser Display	Not Secure	Padlock Symbol

FTP (File Transfer Protocol)

FTP is used for transferring files between computers.

Port Number

21

Features

- Upload files

- Download files
- Rename files
- Delete files
- File sharing

Working

1. Client requests connection.
2. Server accepts connection.
3. Authentication occurs.
4. File transfer begins.

TFTP (Trivial File Transfer Protocol)

TFTP is a simplified version of FTP.

Features

- Uses UDP
- Faster than FTP
- No authentication
- Small file transfers

Port

69

Uses

- Router configuration
- Network booting

Email System

Email allows electronic message exchange over networks.

Main Components:

1. User Agent
2. Mail Server

3. Mail Protocols

User Agent (UA)

Software used by users to send and receive emails.

Examples:

- Gmail
- Outlook
- Thunderbird

Functions:

- Compose mail
 - Send mail
 - Receive mail
 - Read mail
-

SMTP (Simple Mail Transfer Protocol)

SMTP is used for sending emails.

Port

25

Features

- Push protocol
- Sends mail to server
- Server-to-server transfer

Working

Sender → SMTP Server → Receiver SMTP Server

POP3 (Post Office Protocol Version 3)

Used for receiving emails.

Port

110

Features

- Downloads emails
 - Removes mail from server
 - Offline reading possible
-

IMAP (Internet Message Access Protocol)

Used for receiving emails while keeping messages on the server.

Port

143

Features

- Synchronization
- Multi-device access
- Server-based storage

Advantage over POP3

Emails remain available on server.

Difference Between POP3 and IMAP

Feature	POP3	IMAP
Storage	Local Device	Mail Server
Multi-device Access	No	Yes
Synchronization	No	Yes
Internet Required	Only during download	Usually required

DNS (Domain Name System)

DNS converts domain names into IP addresses.

Example

www.google.com

↓

142.250.183.78

Why DNS is Needed

Humans remember names easily.

Computers communicate using IP addresses.

DNS acts like the internet's phonebook.

Port

53

Uses

- Name resolution
 - Domain management
 - Website access
-

DNS Resolution Process

Step 1:

User enters domain name.

Step 2:

DNS query is sent.

Step 3:

DNS server searches records.

Step 4:

IP address returned.

Step 5:

Browser connects to website.

DHCP (Dynamic Host Configuration Protocol)

DHCP automatically assigns IP addresses to devices.

Ports

Server:

67

Client:

68

Information Provided

- IP Address
- Subnet Mask
- Default Gateway
- DNS Server

DHCP Working (DORA Process)

D – Discover

Client broadcasts request.

O – Offer

Server offers IP address.

R – Request

Client requests offered IP.

A – Acknowledge

Server confirms assignment.

Telnet

Telnet provides remote access to another computer.

Port

23

Features

- Remote login
- Command execution

Limitation

Data transmitted in plain text.

Not secure.

SSH (Secure Shell)

SSH is a secure replacement for Telnet.

Port

22

Features

- Encrypted communication
- Remote login
- Secure file transfer

Advantages

- Authentication
 - Confidentiality
 - Integrity
-

SNMP (Simple Network Management Protocol)

Used for monitoring and managing network devices.

Port

161

Devices Managed

- Routers
- Switches
- Servers
- Printers

Functions

- Performance monitoring
- Fault detection

- Configuration management
-

NTP (Network Time Protocol)

Synchronizes clocks across network devices.

Port

123

Uses

- Time synchronization
 - Accurate logging
 - Network coordination
-

Client-Server Architecture

Application layer generally follows a client-server model.

Client

Requests service.

Examples:

- Browser
- Email client

Server

Provides service.

Examples:

- Web server
- Mail server

Example

Browser → Request

Web Server → Response

Peer-to-Peer (P2P) Architecture

In P2P, all devices can act as both client and server.

Features

- Resource sharing
- Decentralized
- No dedicated server

Examples:

- BitTorrent
 - Blockchain networks
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Advantages of Application Layer

- Provides user-friendly services
 - Supports communication between applications
 - Enables web browsing and email
 - Supports file transfer
 - Provides security through HTTPS and SSH
 - Supports remote access
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Disadvantages

- Security threats if not configured properly
 - Application-level attacks possible
 - Dependency on underlying layers
 - Some protocols generate high overhead
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Important Exam Questions

Short Questions

1. Define Application Layer.
2. List functions of the Application Layer.
3. What is DNS?

4. What is DHCP?
5. What is FTP?
6. What is SMTP?
7. Differentiate HTTP and HTTPS.
8. What is SSH?
9. What is Telnet?
10. What is IMAP?

Long Questions

1. Explain Application Layer services.
2. Describe HTTP and HTTPS in detail.
3. Explain DNS working with diagram.
4. Explain DHCP DORA process.
5. Discuss Email architecture and protocols.
6. Compare POP3 and IMAP.
7. Explain FTP working.
8. Compare Client-Server and Peer-to-Peer architecture.

Quick Revision Table

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POP3	Receiving Email	110
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Protocol	Purpose	Port
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DHCP	IP Assignment	67/68
Telnet	Remote Login	23
SSH	Secure Remote Login	22
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NTP	Time Synchronization	123